



MODULE NAME:	MODULE CODE:
SYSTEM ANALYSIS AND DESIGN	SAND6221/p

ASSESSMENT TYPE:	EXAMINATION (PAPER ONLY)
TOTAL MARK ALLOCATION:	120 MARKS
TOTAL HOURS:	2 HOURS (+10 minutes reading time)

INSTRUCTIONS:

1. Please adhere to all instructions in the assessment booklet.
2. Independent work is required.
3. Ten minutes is dedicated to reading time before the start of the assessment. You may make notes on your question paper, but not in your answer sheet. Calculators may not be used during reading time.
4. You may not leave the assessment venue during reading time, or during the first hour or during the last 15 minutes of the assessment.
5. Ensure that your name is on all pieces of paper or books that you will be submitting. Submit all the pages of this assessment's question paper as well as your answer script.
6. Answer all the questions on the answer sheets or in answer booklets provided. The phrase 'END OF PAPER' will appear after the final set question of this assessment.
7. Remember to work at a steady pace so that you are able to complete the assessment within the allocated time. Use the mark allocation as a guideline as to how much time to spend on each section.

Additional instructions:

1. This is a CLOSED BOOK assessment.
2. This assessment has Five Sections. You are required to answer All of these sections.
3. Answer All Questions.
4. Ensure that you have received a special answer booklet (with forms drawn up) for entering your answers. If you have not received an answer booklet, request one from the invigilator before starting the assessment. Answer all questions in the answer booklet provided.

Examination Outcomes

Learning Unit/s Objectives covered in this exam:

At the end of this assessment, students should be able to:

- LU1 - LO1: Describe the purpose of system analysis and design in the development of information systems;
- LU1 - LO2: Describe the characteristics of iterative system development;
- LU1 - LO5: Identify key diagrams used in systems analysis and design;
- LU2 - LO2 Explain the difference between functional and non-functional requirements;
- LU2 - LO3: Identify and Understand different kinds of stakeholders and contributions to requirements definition;
- LU2 - LO6: Apply the user goal technique to identify use cases;
- LU2 - LO11: Develop use case descriptions;
- LU3 - LO2: Explain each major design activity;
- LU3 - LO7: Discuss the impact of visibility and affordance on usability;
- LU4 - LO2: Describe a system development methodology;
- LU4 - LO3: Describe the key features of Agile development;
- LU4 - LO4: Describe the knowledge areas in the Project Management Body of Knowledge (PMBOK);
- LU5 - LO3: Develop design class diagrams;
- LU5 - LO8: Determine when and how to design the database;
- LU5 - LO10: Discuss various types of software tests.

Question 1**(Marks: 15)**

Read the short scenario then answer the questions that follow based on the scenario.

Scenario:

Hire-a-Nanny is a company that allows parents to hire a qualified nanny at short notice. Over the past couple of months, they have noticed that their customer base has grown exponentially, and they have realised the need for an information system to help keep track of bookings and customer interactions.

Q.1.1	In terms of “Hire-a-Nanny”, provide a brief description of what “Systems Analysis” and “Systems Design” entail. Then, <u>explain</u> the role that each of these concepts will play in the development of the new system for “Hire-a-Nanny”.	(6)
Q.1.2	Briefly explain how an iterative development approach can be followed during the development of the “Hire-a-Nanny” system.	(7)
Q.1.3	Identify any two (2) diagrams that will be created during the analysis and design of the “Hire-a-Nanny” system.	(2)

Question 2**(Marks: 35)**

The background information below relates to the short scenario provided at the beginning of **Question 1**. Use the information below to answer the questions in this section.

Parents do not always have the time to book a Nanny during office hours and as such the system to be developed for “Hire-a-Nanny” should be available 24/7. This means that a parent should be able to make a booking or track the confirmation of a booking at a time that is suitable to them. As the system will store the personal details as well as payment information of parents wishing to hire a nanny, the data in the database should be password protected.

For the management of “Hire-a-Nanny”, the new system would bring about greater efficiency. For the Nannies, it will provide a controlled environment in which they can offer their expertise according to their availability and will enable them to manage their time more effectively. For parents it will ensure peace-of-mind knowing that they leave their children in capable hands.

Q.2.1	Explain the difference between functional and non-functional requirements. Substantiate your answer by providing two examples of functional requirements and two (2) examples of non-functional requirements from the information provided at the beginning of Question 2 .	(10)
Q.2.2	From the scenario, identify any three (3) stakeholders. Then provide an example of one piece of information that each of these stakeholders can provide you as a Systems Analyst with that could potentially assist you with clarifying the requirements for the system to be developed.	(9)
Q.2.3	Consider all information provided about the “Hire-a-Nanny” system, then show how the user goal technique for identifying use cases can applied in order to identify a use case for the new system.	(10)
Q.2.4	Identify any six (6) pieces of data that will form part of a use case description.	(6)

Question 3 (Marks: 20)

Answer all questions in this section.

Q.3.1	Describe the five (5) activities that forms part of Systems Design.	(15)
Q.3.2	Given what you know about the “Hire-a-Nanny” system, describe the importance of visibility and affordance on the usability of the information system.	(5)

Question 4 (Marks: 25)

Answer all questions in this section.

Q.4.1	Briefly discuss the concept of a system development methodology. As part of your answer provide the following: a) A general description of what a system development methodology is; b) A description of each component of a system development methodology and c) Three examples of each component of a system development methodology.	(17)
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Q.4.2	List the four (4) basic values of the Agile Development approach.	(4)
Q.4.3	Identify and describe any two (2) knowledge areas within the Project Management Body of Knowledge (PMBOK).	(4)

Question 5**(Marks: 25)**

Answer all questions in this section.

Q.5.1	When a nanny registers her services with “Hire-a-Nanny” she needs to provide her name, surname, age, qualifications and availability. Create a design class diagram that will accurately represent a class called ‘Nanny’. The design class diagram must show the following: a) Five attributes with visibility and data type indicated; b) Any five (5) relevant method signatures.	(16)
Q.5.2	Provide a brief discussion as to the factors that influence when a database will be designed.	(6)
Q.5.3	List any three (3) software tests that can be performed.	(3)

END OF PAPER